Evaluating the Role of Language in Small Exact Number Concepts

Jenna Croteau[1,2] (jmargarites@umass.edu), Özge Bakay[1,3], Brian Dillon[1], Joonkoo Park[2]

[1]*Linguistics, University of Massachusetts, Amherst*

[2]*Psychological and Brain Sciences, University of Massachusetts, Amherst*

[3]*Linguistics, University of Michigan, Ann Arbor*

Introduction

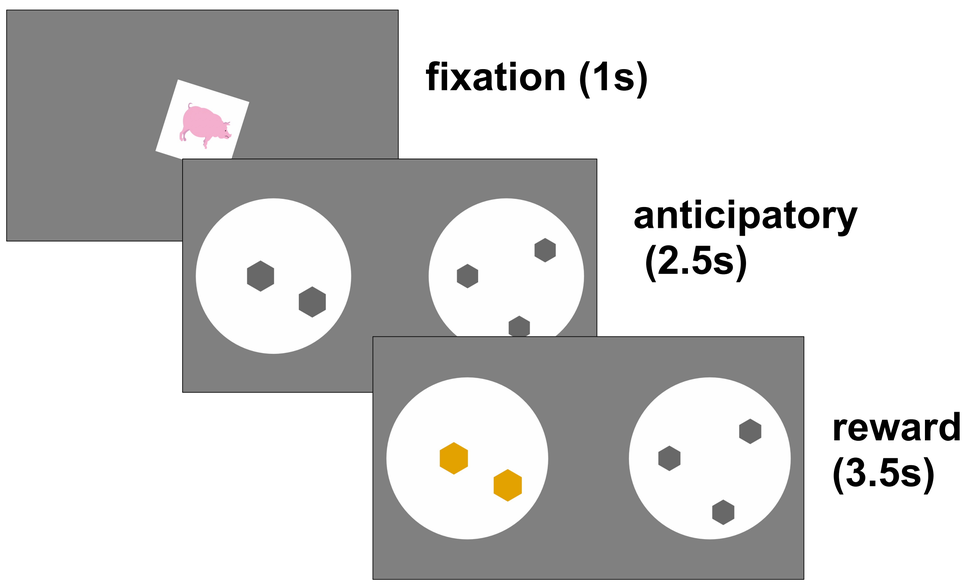
- Empirical evidence suggests language is related to the development of large exact number concepts [1,2]
- Theoretical proposals suggest that language may also be related to the development of small exact number concepts [3,4]; however empirical support is mixed [1,4, 5]

RESEARCH QUESTION: Is language required for exact small number concepts?

Methods

Anticipatory Looking Paradigm

36 trials (fixation, anticipatory period, reward)

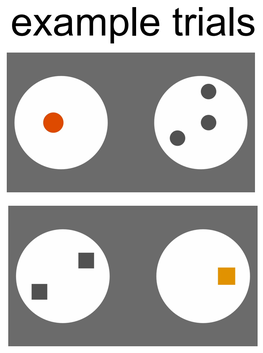


Example trial where Target = 2 and Target side = left

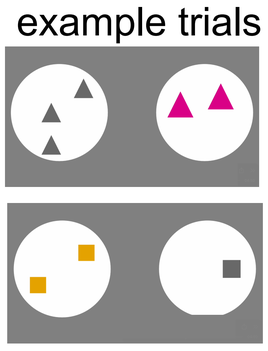
Experimental Conditions

Participants were randomly assigned to Target=1, 2, or 3. Trial stimuli were identical across experiments.

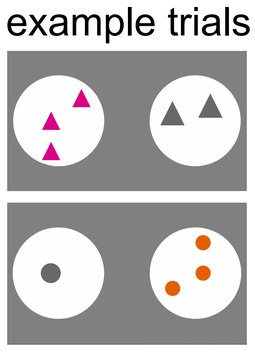
Target=1
example trials



Target=2
example trials



Target=3
example trials

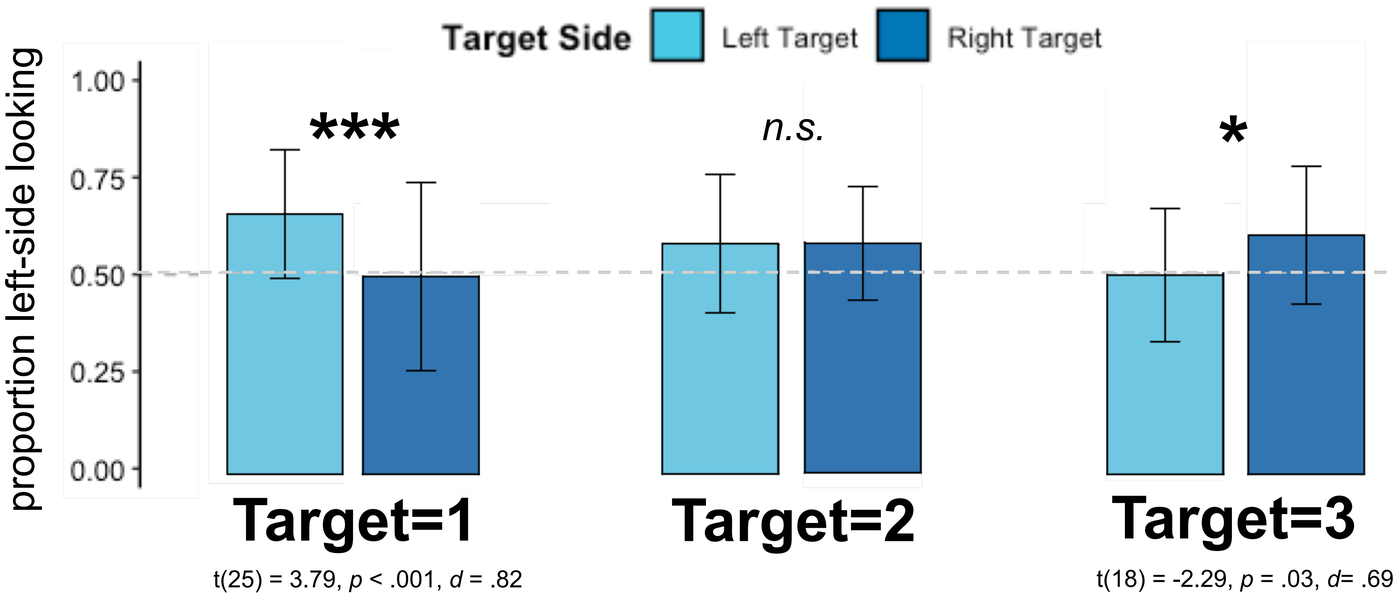


Adult participants were randomly assigned to a **control** or **verbal shadowing** condition

Results

Infant Data

(n=72; M_{age}=17.3 months, range=14-23 months; between-subjects design)

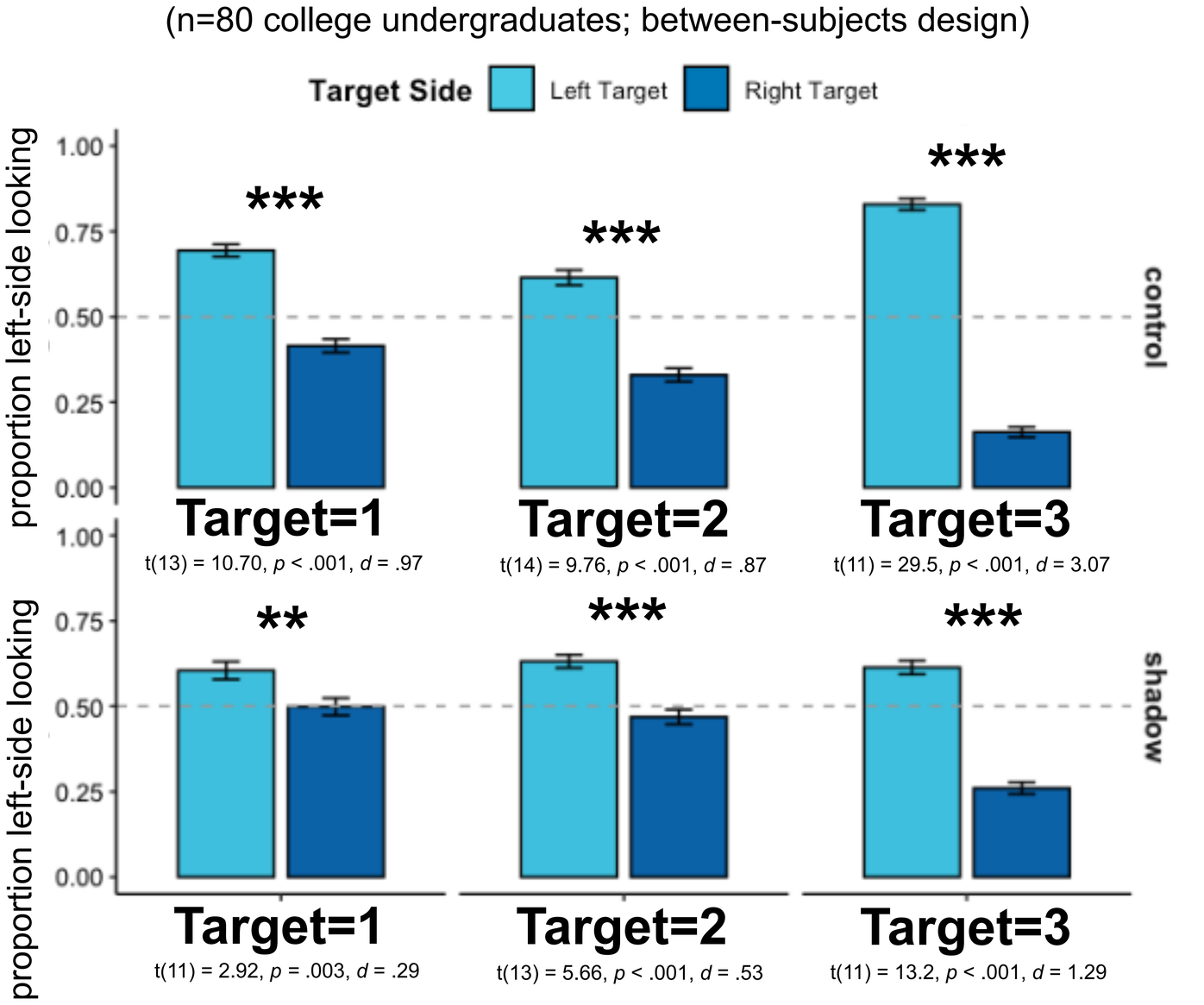


t(25) = 3.79, p < .001, d = .82

t(18) = -2.29, p = .03, d= .69

Adult Data

(n=80 college undergraduates; between-subjects design)



t(13) = 10.70, p < .001, d = .97

t(14) = 9.76, p < .001, d = .87

t(11) = 29.5, p < .001, d = 3.07

t(11) = 2.92, p = .003, d = .29

t(13) = 5.66, p < .001, d = .53

t(11) = 13.2, p < .001, d = 1.29

* p <.05, ** p<.01, *** p<.001

Discussion

- Infants show “successful” anticipatory looking for target=1 but not target=2 or =3
- Adults show ”successful” anticipatory looking for target=1, 2, and 3 regardless of verbal shadowing or control condition assignment
- A significant difference the control condition and verbal shadowing condition suggests shadowing made the anticipatory looking task more difficult but not impossible
- Results are most compatible with the idea that **language is not required for exact small number concepts** (at least in adults)
- In future work we hope to explore individual variation; the significant result in the adult target=1 verbal shadowing condition is driven by 4 participants

Acknowledgements

Thank you to Diana Mendel, Marina Morocos, Carina Chan, Ally Kim, Deb Foucault, & Shannon Ganser for assistance with coding the infant looking time videos. Thank you to Saf Costa, Kirthi Sivakumar, and Tony Jean-Baptiste for their assistance with adult data collection. This project was partially funded by NSF Award BCS #2213770 and the Armstrong Graduate Student Award (2026). And finally, a very special thank you to the babies, families, and college undergraduates for their participation in this work.

BCS #2213770

Children Helping Science

Cognitive & Developmental Neuroscience Lab

1 Frank, M. C., Fedorenko, E., Lai, P., Saxe, R., & Gibson, E. (2012). Verbal interference suppresses exact numerical representation. *Cognitive psychology*, *64*(1-2), 74-92; 2 Pitt, B., Gibson, E., & Piantadosi, S. T. (2022). Exact number concepts are limited to the verbal count range. *Psychological Science*, *33*(3), 371-381. 3 Barner, D. (2017). Language, procedures, and the non-perceptual origin of number word meanings. Journal of child language, 44(3), 553-590; 4 Le Corre, M., Li, P., Huang, B. H., Jia, G., & Carey, S. (2016). Numerical morphology supports early number word learning: Evidence from a comparison of young Mandarin and English learners. *Cognitive psychology*, *88*, 162-186; 5 Frank, M. C., Everett, D. L., Fedorenko, E., & Gibson, E. (2008). Number as a cognitive technology: Evidence from Pirahã language and cognition. *Cognition*, *108*(3), 819-824.